ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** 2 m/s squared upward	R = 720 N: m(g + a).
3	**(d)** the pull of the book on the Earth	Third-law pair.
4	**(d)** 4 m/s squared	F/m.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 2 m/s squared downward	m(g - a) = 480.
7	**(b)** reducing the friction by rollers	Reduce friction.
8	**(a)** 200 N	0.1 x 20 / 0.01.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(c)** 10 m/s squared downward	Only gravity acts.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	a = -2 m/s squared (1); t = 5 s, s = 25 m (1).	1+1
12	Normal force from the ground balances the weight (1); Net force zero (1).	1+1
13	Net force zero at constant velocity (1); Hence pull equals friction (1).	1+1
14	Same direction: 14/2 = 7 m/s squared (1); Opposite: 4/2 = 2 m/s squared (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	a = 4 m/s squared (1); F = 6000 N, s = 50 m (1); Weight = 15000 N; force is 0.4 of weight (1).	1+1+1
16	a = v squared/2s = 2 x 10^6 m/s squared (1); F = 0.01 x 2 x 10^6 = 20000 N (1); t = v/a = 0.0002 s (1).	1+1+1
17	F = 10/0.01 = 1000 N (1); Third law: nail exerts 1000 N on the hammer (1); Force on the nail 1000 N (1).	1+1+1
	OR: a = 5 m/s squared (1); v = 20 m/s, change in momentum 120 kg m/s (1); s = 40 m (1).	1+1+1
18	At 40 N: net force zero, constant velocity or rest (1); At 60 N: net 20 N (1); Box accelerates in the direction of the push (1).	1+1+1
	OR: Upward acceleration: more (1); Constant velocity: same (1); Downward acceleration: less; free fall: zero (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	a = 3 m/s squared (1); v = 15 m/s (1); s = 37.5 m (1); p = 120 kg m/s (1); F = change in p/t = 24 N (1).	1+1+1+1+1
	OR: Change = 0.16 x 70 = 11.2 kg m/s (1); Impulse = 11.2 N s (1); F = 560 N (1); Doubling time halves force (1); Same impulse (1).	1+1+1+1+1
20	(a) 70 kg (1); (b) 91 kg (1); (c) 49 kg (1); (d) 0 kg (1); Reading = m(g plus or minus a)/g (1).	1+1+1+1+1
	OR: Net = sqrt(900 + 1600) = 50 N (1); a = 10 m/s squared (1); tan = 40/30, about 53 degrees north of east (1); Second law (1); Vector addition (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** the same for both	F = change in p/t.
2	**(c)** 10 N	20/2.
3	**(c)** A is true but R is false.	A is true; R is false: mass does not depend on gravity.
4	**(c)** 300 N	m a + f = 200 + 100.
5	**(b)** 1 : 2	v = Ft/m.
6	**(b)** 10 N s	Ft.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(c)** 6 N	a = 2; force on the 3 kg block = 6.
9	**(d)** 30 kg m/s	Impulse = Ft.
10	**(c)** 8 m/s squared	a = 2F/(m/2).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Constant velocity: 700 N (1); Accelerating down: 70 x 7 = 490 N (1).	1+1
12	Impulse 1 N s (1); v = 2 m/s (1).	1+1
13	F = rate of change of momentum (1); = 5 x 2000 = 10000 N (1).	1+1
14	Body keeps the forward velocity of the train (1); Feet stop on the ground, upper body continues and he falls (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	T = 800 x 11.5 = 9200 N (1); R = 60 x 11.5 = 690 N (1); Weight of person 600 N (1).	1+1+1
16	Limiting static friction = 250 N (1); Net force when moving = 100 N (1); a = 2 m/s squared (1).	1+1+1
17	a = 3 m/s squared (1); v = 15 m/s (1); From rest (1).	1+1+1
	OR: a = 5 m/s squared for 2 s: v = 10 m/s (1); Then constant 10 m/s (1); s = 10 + 30 = 40 m (1).	1+1+1
18	Same bullet momentum (1); Recoil speed inversely proportional to mass (1); Heavier gun recoils less (ratio 4 : 1) (1).	1+1+1
	OR: a = 2 m/s squared (1); v = 8 m/s (1); Change in momentum = 12 kg m/s (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Change in momentum 5 kg m/s (1); F1 = 100 N (1); F2 = 20 N (1); Same change, longer time gives smaller force (1); Used in cricket catching (1).	1+1+1+1+1
	OR: First law: ball at rest stays until force acts (1); Impulse = 100 x 0.05 = 5 N s (1); v = 5/0.4 = 12.5 m/s (1); Third law: the foot feels an equal and opposite force (1); Friction and air resistance finally stop the ball (1).	1+1+1+1+1
20	Change in momentum 900 kg m/s for passenger (1); With airbag (0.3 s): 3000 N (1); Without (0.1 s): 9000 N (1); Airbag reduces the force to one-third (1); F = change in momentum/time (1).	1+1+1+1+1
	OR: Engine force overcomes resistance (1); Net force gives acceleration (1); Inertia in braking (1); Seat belt, airbag, crumple zone (1); Speed limits reduce momentum (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **How Forces Affect Motion** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** moves away from the shore	Third law.
3	**(a)** 500 N	m a on the trailer.
4	**(a)** 4 m	v squared/2a.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(d)** 7 m/s	a = 2; v = 3 + 4.
7	**(d)** a1 a2/(a1 + a2)	F/(m1 + m2) with m = F/a.
8	**(b)** 10 m/s	a = 2.5.
9	**(d)** A is false but R is true.	A is false: all bodies fall with the same acceleration; R is true but does not support A.
10	**(d)** 10 N	m a.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Friction at the axle and air drag (1); Unbalanced force slows it (1).	1+1
12	v = 10 m/s (1); F = 600/0.2 = 3000 N (1).	1+1
13	a = -2 m/s squared; F = 4 N (1); s = 9 m (1).	1+1
14	100 N in the opposite direction (1); Wall does not move due to ground friction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Air resistance increases with speed (1); At terminal velocity net force zero (1); Resistance = weight = 700 N (1).	1+1+1
16	Change = 0.15 x 70 = 10.5 kg m/s (1); F = 1050 N (1); Impulse 10.5 N s (1).	1+1+1
17	a1 = 6 m/s squared (1); a2 = 2 m/s squared (1); More inertia, less acceleration (1).	1+1+1
	OR: (a) 50 N (1); (b) 60 N (1); (c) 40 N (1).	1+1+1
18	The stone moves along the tangent (1); No force acts to change its direction (1); Inertia of motion (1).	1+1+1
	OR: F = 3 N (1); u = 4 m/s (1); v = 8 m/s (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Momentum per bullet 10 kg m/s (1); 10 bullets per s: 100 kg m/s per s (1); F = 100 N (1); Third law: the bullets push the gun back (1); Force to hold = 100 N (1).	1+1+1+1+1
	OR: a = 5/(2.5) = 2 m/s squared (1); For the weight: 5 - T = 0.5 x 2 (1); T = 4 N (1); Trolley: T = 2 x 2 = 4 N (1); Check consistent (1).	1+1+1+1+1
20	Net force 40 N (1); a = 4 m/s squared (1); v = 20 m/s (1); s = 50 m (1); p = 200 kg m/s (1).	1+1+1+1+1
	OR: a = 5 m/s squared (1); s = 40 m, t = 4 s (1); Double speed: s = 160 m, t = 8 s (1); Distance four times, time twice (1); Important for speed limits (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.